

Bonus: Operational Amplifiers

Ideal op-amp rules, inverting, non-inverting, follower, summing, difference, integrator, saturation

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 7 worked examples, and 6 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §6.1–6.6

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

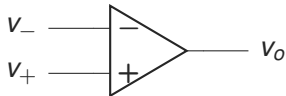
Contents

PART 1: Lecture notes	3–22
What an op-amp is, the two golden rules	3–5
Inverting amplifier	6–8
Non-inverting amplifier, voltage follower (buffer)	9–12
Summing and difference amplifiers	13–15
Integrator	16–19
Saturation and real limits	20–20
Traps and summary	21–22
PART 2: Final check (6 practice problems)	23–29
PART 3: Answer sheet (full solutions)	30–36

Note: op-amps are listed in the course description but are examined less often than the main topics. Learn this deck after the main weeks.

What an op-amp is

A very high-gain amplifier with two inputs:



$$v_o = A(v_+ - v_-), \quad A \approx 10^5 \text{ to } 10^6$$

Used almost always with **negative feedback**: part of the output fed back to the $-$ input. The feedback resistors, not A , then set the gain.

The two golden rules (ideal op-amp with negative feedback)

1. **No current into the inputs:** $i_+ = i_- = 0$.
2. **The inputs are at the same voltage:** $v_+ = v_-$ (“virtual short”).

Why the golden rules are true

Rule 1 (no input current): real op-amp inputs have huge resistance ($M\Omega$ to $T\Omega$), so the current into them is tiny compared with the other currents in the circuit. Treat it as zero.

Rule 2 ($v_+ = v_-$): the output is $v_o = A(v_+ - v_-)$ with $A \approx 10^5$ or more. A normal output is a few volts at most, so

$$v_+ - v_- = \frac{v_o}{A} \approx \frac{10 \text{ V}}{10^5} = 0.1 \text{ mV} \approx 0$$

Why feedback is needed: with negative feedback, if v_o rises too far it pulls v_- up, which shrinks $v_+ - v_-$ and brings v_o back down. The circuit settles exactly where $v_+ \approx v_-$. Without feedback, the tiny difference is multiplied by 10^5 and the output slams into a supply rail.

The recipe for any op-amp circuit

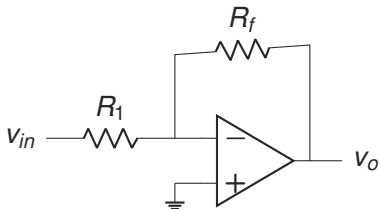
- 1 Find v_+ (usually easy: ground or an input).
- 2 Set $v_- = v_+$ (rule 2).
- 3 Write KCL at the $-$ node, with **zero** current into the op-amp (rule 1).
- 4 Solve for v_o .

Trap

Never write KCL at the output node: the op-amp supplies whatever output current it needs. And the rules only hold with **negative** feedback (output to the $-$ input).

Inverting amplifier

[NOT on formula sheet: memorize]



$$v_o = -\frac{R_f}{R_1} v_{in}$$

$v_+ = 0$, so $v_- = 0$ (“virtual ground”).

$$\text{KCL at } -: \frac{v_{in} - 0}{R_1} = \frac{0 - v_o}{R_f}.$$

The minus sign means the output is **inverted** (180° out of phase for AC).
The input resistance seen by the source is just R_1 .

Try it: Example 1, inverting amplifier

An inverting amplifier has $R_1 = 2 \text{ k}\Omega$, $R_f = 10 \text{ k}\Omega$ and $v_{in} = 0.5 \text{ V}$.

Find the gain and v_o .

your work here ↓

Example 1: solution

Formula inverting: $v_o = -\frac{R_f}{R_1} v_{in}$

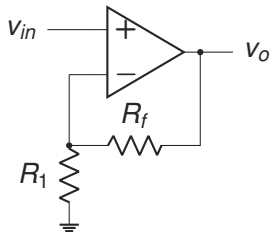
Gain $-10/2 = -5$

Output $-5 \times 0.5 = -2.5 \text{ V}$ (opposite sign to the input)

A real op-amp with $A = 10^6$ gives -2.499985 V : the ideal rules are extremely accurate.

Non-inverting amplifier

[NOT on formula sheet: memorize]



$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_{in}$$

$v_- = v_+ = v_{in}$, and R_1 , R_f form a divider from v_o down to v_- .

Voltage follower (buffer)

$R_f = 0$ (wire) and $R_1 = \infty$ (removed): $v_o = v_{in}$, gain 1. It draws no current from the source but can drive a load: it **isolates** stages (week 1's loaded-divider problem disappears).

Deriving the non-inverting gain

Rule 2 $v_- = v_+ = v_{in}$

Rule 1 no current into the $-$ input, so the same current flows through R_f and then R_1 to ground

Divider R_f and R_1 form a divider from v_o down to ground, with v_- in the middle: $v_- = v_o \frac{R_1}{R_1 + R_f}$

Combine $v_{in} = v_o \frac{R_1}{R_1 + R_f} \Rightarrow v_o = \frac{R_1 + R_f}{R_1} v_{in} = \left(1 + \frac{R_f}{R_1}\right) v_{in}$

The gain is always at least 1, and the output has the same sign as the input (not inverted). The source sees an open circuit (no input current), so it isn't loaded at all.

Try it: Example 2, non-inverting and buffer

- (a) Non-inverting amplifier with $R_1 = 1 \text{ k}\Omega$, $R_f = 9 \text{ k}\Omega$, $v_{in} = 0.3 \text{ V}$. Find v_o .
- (b) A 10 V source feeds a divider of two 10 k Ω resistors. A 1 k Ω load is connected across the bottom resistor, first directly, then through a buffer. Load voltage in each case?

your work here ↓

Example 2: solution

(a) non-inverting: $1 + R_f/R_1 = 1 + 9/1 = 10$:

$$v_o = 10 \times 0.3 = 3 \text{ V}$$

(b) **direct** the 1 k load sits in parallel with the bottom 10 k:

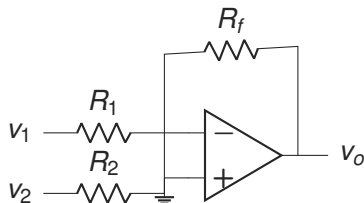
$$10 \parallel 1 = 0.909 \text{ k}$$

loaded divider: $10 \cdot \frac{0.909}{10 + 0.909} = 0.83 \text{ V}$ (the load drags the voltage way down)

(b) **buffer** the follower draws no current, so the divider stays unloaded at 5 V; the buffer copies it and drives the load from its own output: 5 V

Summing amplifier

[NOT on formula sheet: memorize]



$$v_o = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \dots \right)$$

The currents from each input add at the virtual ground.

Difference amplifier

With R_1 into the $-$ input, R_2 as feedback, and a matching R_1/R_2 divider on the $+$ input:

$$v_o = \frac{R_2}{R_1} (v_2 - v_1) \quad (v_2 \text{ on the } + \text{ side, } v_1 \text{ on the } - \text{ side})$$

Try it: Example 3, summing and difference

- (a) Summing amplifier: $R_f = 10\text{ k}$, $R_1 = 10\text{ k}$ with $v_1 = 1\text{ V}$, $R_2 = 5\text{ k}$ with $v_2 = 0.5\text{ V}$. Find v_o .
- (b) Difference amplifier with $R_2/R_1 = 4$, $v_2 = 1.5\text{ V}$, $v_1 = 1\text{ V}$. Find v_o .

your work here ↓

Example 3: solution

(a) idea at the virtual ground each input's current is v_k/R_k ; they add and all flow through R_f

$$(a) \quad v_o = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} \right) = -10 \left(\frac{1}{10} + \frac{0.5}{5} \right) = -10(0.2) =$$

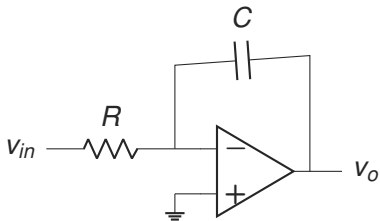
-2 V

$$(b) \quad \text{difference amplifier: } v_o = \frac{R_2}{R_1}(v_2 - v_1) = 4(1.5 - 1) =$$

2 V

Integrator

[NOT on formula sheet: memorize]



$$v_o(t) = -\frac{1}{RC} \int_0^t v_{in} dt + v_o(0)$$

Replace R_f with a capacitor. A constant input gives a **ramp** output.

Example 4

$R = 10 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$ ($RC = 10 \text{ ms}$), $v_{in} = 1 \text{ V}$ constant, $v_o(0) = 0$:
slope $= -1/0.01 = -100 \text{ V/s}$, after 50 ms: $v_o = -5 \text{ V}$.

Deriving the integrator

Virtual ground $v_+ = 0$, so $v_- = 0$

Input current through R : $i = (v_{in} - 0)/R = v_{in}/R$

Into C no current enters the op-amp, so all of i flows into the capacitor

Capacitor voltage across C is $v_- - v_o = -v_o$, and $i = C \frac{d(-v_o)}{dt}$

Combine $\frac{v_{in}}{R} = -C \frac{dv_o}{dt} \Rightarrow v_o(t) = -\frac{1}{RC} \int_0^t v_{in} dt + v_o(0)$

A constant input pushes a constant current into C, so its voltage (and v_o) changes at a steady rate: a ramp.

Differentiator

[NOT on formula sheet: memorize]

Swap the R and C of the integrator: C at the input, R as feedback. The virtual ground holds $v_- = 0$, so the input capacitor's current is $i = C dv_{in}/dt$, and all of it flows through R:

$$0 - v_o = R C \frac{dv_{in}}{dt} \quad \Rightarrow \quad v_o = -RC \frac{dv_{in}}{dt}$$

Example: $RC = 10$ ms, input ramp rising 0.5 V/ms (500 V/s):

$v_o = -0.01 \times 500 = -5$ V, constant while the ramp lasts.

Caution: a differentiator amplifies fast changes, so it amplifies noise. In practice it's rarely used alone.

Active low-pass filter, and why stages chain

[NOT on formula sheet: memorize]

Put a capacitor C in parallel with R_f in an inverting amplifier.

At low frequency the capacitor is an open circuit, so the gain is the usual $-R_f/R_1$.

At high frequency the capacitor shorts R_f , and the gain falls toward zero.

The switch-over (cutoff) happens where $|Z_C| = R_f$: $f_c = \frac{1}{2\pi R_f C}$.

Example: $R_1 = 1 \text{ k}$, $R_f = 10 \text{ k}$, $C = 15.9 \text{ nF}$: gain -10 ,

$$f_c = \frac{1}{2\pi(10^4)(15.9 \times 10^{-9})} = 1.00 \text{ kHz}$$

Input impedance of an inverting stage $= R_1$; output impedance of an ideal op-amp $= 0$. That's why stages can be chained without loading each other.

Saturation and real limits

Saturation

The output can't exceed the supply rails (about $\pm V_{CC}$). If the formula asks for more, the output **clips** at the rail.

Example 5

Inverting gain -10 , supplies $\pm 12\text{ V}$, $v_{in} = 2\text{ V}$: the formula says -20 V , but the output is -12 V (saturated).

Other real limits (§6.6): finite gain, input offset voltage, finite bandwidth (gain drops at high frequency), slew rate (the output can only change so fast), and maximum output current.

Traps

Check these before you box an answer

- The golden rules need negative feedback.
- Zero current into the inputs, but **not** out of the output.
- Inverting: $-R_f/R_1$. Non-inverting: $1 + R_f/R_1$. Don't mix them.
- Check the rails: the answer can't be bigger than the supply.
- Integrator: $-1/RC$, and don't forget the initial output voltage.

Summary

Golden rules

$i_+ = i_- = 0$, $v_+ = v_-$ (with negative feedback)

Inverting

$$v_o = -(R_f/R_1)v_{in}$$

Non-inverting

$v_o = (1 + R_f/R_1)v_{in}$; follower: gain 1, buffers stages

Summing

$$v_o = -R_f \sum v_k/R_k$$

Difference

$$v_o = (R_2/R_1)(v_2 - v_1)$$

Integrator

$$v_o = -\frac{1}{RC} \int v_{in} dt$$

Limits

output clips at the rails

Final check

6 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: inverting

$R_1 = 4.7 \text{ k}\Omega$, $R_f = 47 \text{ k}\Omega$, $v_{in} = 0.2 \text{ V}$. Find v_o .

your work here ↓

P2: non-inverting

$R_1 = 5 \text{ k}\Omega$, $R_f = 15 \text{ k}\Omega$, $v_{in} = 0.5 \text{ V}$. Find v_o .

your work here ↓

P3: summing

$R_f = 20\text{ k}$. Inputs: 0.3 V through 10 k, 0.2 V through 4 k, 1.0 V through 20 k.
Find v_o .

your work here ↓

P4: difference

$R_1 = 2 \text{ k}$, $R_2 = 10 \text{ k}$, $v_2 = 2.2 \text{ V}$, $v_1 = 2.0 \text{ V}$. Find v_o .

your work here ↓

P5: saturation

An inverting amplifier with gain -8 runs on ± 15 V. What is the largest input magnitude before it saturates?

your work here ↓

P6: integrator

$R = 10 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$, $v_o(0) = 0$. The input is 0.5 V for 20 ms. Output at 20 ms?

your work here ↓

Answer sheet

Full worked solutions to P1–P6.

P1: solution

Gain inverting: $-R_f/R_1 = -47/4.7 = -10$

Output $-10 \times 0.2 = -2 \text{ V}$

P2: solution

Gain non-inverting: $1 + R_f/R_1 = 1 + 15/5 = 4$

Output $4 \times 0.5 = 2 \text{ V}$

P3: solution

Currents each input into the virtual ground: $\frac{0.3}{10} = 0.03$, $\frac{0.2}{4} = 0.05$,
 $\frac{1.0}{20} = 0.05$ mA

Sum 0.13 mA, all through $R_f = 20$ k

Output $v_o = -20 \times 0.13 = -2.6$ V

P4: solution

Formula $v_o = \frac{R_2}{R_1}(v_2 - v_1)$

Result $\frac{10}{2}(2.2 - 2.0) = 5 \times 0.2 = 1 \text{ V}$ (only the *difference* is amplified)

P5: solution

Limit the output can't exceed the 15 V rail: $|v_o| = 8|v_{in}| \leq 15$

Result $|v_{in}| \leq 1.875 \text{ V}$; beyond that the output clips at $\pm 15 \text{ V}$

P6: solution

Rate constant input: v_o changes at $-\frac{v_{in}}{RC} = -\frac{0.5}{0.01} = -50 \text{ V/s}$

Output starting from 0: $-50 \times 0.02 = -1 \text{ V}$